

Thesis Proposal – Reinforcement Learning Environment for Collectible Card Game Battling



TECHNISCHE
UNIVERSITÄT
DARMSTADT

Tim Unverzagt
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1 Introduction

Games have long served AI research as a treasure trove of simulations providing varied challenges with relatively well-established rules, tractable scopes of required skills and abundance of human expertise. After deep-learning Monte-Carlo-Tree-Search approaches achieved unambiguous super-human play in go (and similar symmetric perfect information games with static action and state spaces), research into structurally complex games has intensified. In the last decade collectible card games (CCG) have been the focus of various works due to several such structural complexities:

1. partial information
2. dynamic action space
3. enormous state space
4. complex rules defined in natural language
5. non deterministic game setup (decks)
6. potentially non deterministic state transitions
7. structurally different but connected game sections:
 - a) Drafting
 - b) Deckbuilding
 - c) Battling

While several teams have shown that, in simple CCGs, self learned agents can outperform expert-crafted classical approaches during Drafting few works exist on deep-learning for Battling. From personal experience in several CCGs we believe that, in any sufficiently complex CCG, Battling, Drafting and Deck-Building are linked strongly enough that comprehensive research on either has to ultimately involve the other. Research on Battling thus not only provides a new set of challenges but also enables more informed analysis of previous work on Drafting and consequently is essential for the development of a universal CCG-playing agent.

Current implementations of CCG engines are usually high-fidelity, resource intense implementations intended for human use (or sometimes classical AI) without a proper API for AI training. A notable exception exists in LOCM (see. ref) for which an openai gym API, capable of running multiple games a second, is implemented. While fundamental to research on CCG AI in the last decade LOCM eschews multiple sources of structural complexity to improve accessibility for various self learning approaches and execution speed. It consequently also deviates from environments where human expertise is most abundant, namely preexisting CCGs. We believe that structural complexity and human expertise are essential to improve CCG agents in the long term, thus we propose to implement an CCG environment which offers an efficiently computable, simplified (but congruent) version of a preexisting CCG. The focus of the proposed implementation shall be on computationally inexpensive sources of structural complexity as well as expandability to enable a gradual approach to the reference CCG. Due to its high-complexity even among CCGs, and its wide spread among players and researchers alike, we choose Magic: The Gathering as our reference CCG.

2 State of Research

3 Method
